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***B.Tech. Degree I Semester Regular/Supplementary and
II Semester Supplementary Examination March 2022***

CS/EC/IT 19-200-0101B & CE/EE/ME/SE 19-200-0201A

CALCULUS

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A
(Answer *ALL* questions)

(8 × 3 = 24)

- I. (a) Solve $xy \frac{dy}{dx} = 1 + x + y + xy$.
- (b) Solve $(D^2 + 3D + 2)y = e^{8x}$.
- (c) If $u = \log(e^x + e^y)$; prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 1$.
- (d) If $u = \frac{xy}{x+y}$; prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = u$.
- (e) Find the area of the Lemniscate $r^2 = a^2 \cos(2\theta)$
- (f) Evaluate $\int_0^1 \int_0^{1-x} \int_0^{1-x^2-y^2} dz dy dx$.
- (g) Find the unit vector normal to $xy^2z^3 = 1$ at (1,1,1).
- (h) Prove that (i) $\text{div } \vec{r} = 3$
(ii) $\text{curl } \vec{r} = 0$

PART B

(4 × 12 = 48)

- II. (a) Solve $(1 + y^2)dx = (\tan^{-1} y - x)dy$.
- (b) Solve $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = \cos(\log x)$.
- OR**
- III. (a) Solve $(D^2 - 1)y = x \sin(3x) + \cos x$.
- (b) Solve $\frac{d^2 y}{dx^2} - \frac{6}{x^2} y = x \log x$.
- IV. (a) Verify Euler's theorem for the function $u = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)$.
- (b) If $u = f(y-z, z-x, x-y)$; prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.

OR

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- V. (a) If $x = u - v$ and $y = uv$ and if z is a function of x and y , show that
- $$(u + v) \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}.$$
- (b) The work that must be done in propel a ship of displacement D for a distance S in time t is proportional to $\frac{S^2 D^{3/2}}{t^2}$. Estimate roughly the percentage change in the work when the distance is increased by 1%, the time is diminished by 1% and the displacement of the ship is diminished by 3%.
- VI. (a) Find the surface area of the solid generated by the revolution of the astroid $x^{2/3} + y^{2/3} = a^{2/3}$ about the x -axis.
- (b) Find the volume of a sphere of radius a .
- OR
- VII. (a) Find the area between the cardioids $r = a(1 + \cos \theta)$ and $r = a(1 - \cos \theta)$.
- (b) Change the order of integration and then evaluate the integral
- $$I = \int_0^1 \int_{x=0}^{x=1-y} (x^2 + y^2) dx dy.$$
- VIII. (a) Find the directional derivative of the function $2xy + z^2$ in the direction of $i + 2j + 2k$ at the point $(1, -1, 3)$.
- (b) Show that $\vec{F} = (2xy + z^3)i + x^2j + 3xz^2k$ is a conservative force field. Find the scalar potential.
- OR
- IX. (a) Verity Stoke's theorem for the function $\vec{F} = x^2i + xyj$ integrated around the square in the plane $z = 0$ whose sides are along the lines $x = 0$; $y = 0$; $x = a$; $y = a$.
- (b) If $\vec{F} = (2x^2 - 3z)i - 2xyj - 4xk$, evaluate $\iiint_V \nabla \cdot \vec{F} dV$, where V is bounded by the planes $x = 0$; $y = 0$; $z = 0$ and $2x + 2y + z = 4$.
